Octagonal House Design and Use of Bamboo for Temperature Reduction in Houses

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1. Introduction

Based on an article written by Kompas (2022), South Tangerang, Banten has the highest increase of temperature in the Java island where it reaches an astonishing +0,46^{O} C per decade. The problem with temperature increasing throughout the years causes air conditioners to be one of the most common appliances found in a variety of buildings. In the future, it is predicted that air conditioners would become a secondary need, causing the demand for air conditioners to increase 3 fold by 2050. However despite the benefits, air conditioners make up around 10% of the global electricity consumption and also releases CFC (Chlorofluorocarbon), which is a type of gas that contributes to global warming (euronews.green). Currently, each household has an average of 56% of their electricity bill originating from air conditioners due to their estimated use of 10 hours per day (daikin, 2021).

Due to these issues, a solution to create an eco-friendly alternative for air conditioners is desperately needed in order to stop contributions towards global warming and harming the environment. Hence, our team is attempting to create a house prototype that could exploit the use of shapes and materials (bamboo) to increase the effectiveness of air circulation for ventilation, as well as reducing electricity consumption in houses.

2. Research Methodology

In this experiment, we are comparing the air temperature and water temperature in 2 different houses with different designs. Our independent variable is a house model which is made to increase the effectiveness of air circulation which is in description as follows:

Bamboo ventilated house (independent variable):

- Eight sided/octagonal structure with tilted bamboo walls (30º) and a fan on the centre of the octagonal pyramid roof.

Normal house (*control group*)

- This structure is made based on the common shapes of houses with rectangular walls and a triangular roof.

Dependent variables measured:

- Air temperature (in & out of the houses)
- Water temperature in a beaker (in & out of the houses; representation of a human)

To ensure a valid result, we have some controlled variables which include the area for the base of the house (60 x 60 x 60), construction materials (excluding bamboo), location of experiment), and the paint color for the walls (white). This experiment will be in 2 sets with 3 repetitions each set. Each set will be in the total span of 30 minutes where the data will be collected every 10 minutes. The first experiment will not use the fan, whilst the second experiment will use the fan in order to make the air circulation more effective.

3. Results and Analysis

The following tables and figures are the acquired results from the experiments.

Figure 1.1

Average air and water temperature without the application of a fan in the experimental house

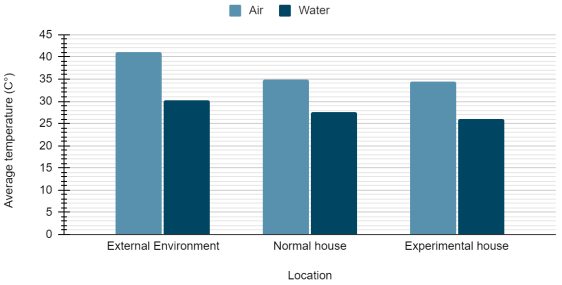


Table 1.1 Result of temperatures for experiment 1

Experiment 1.

Repetitions	Temperature (C°)					
	External Environment		Normal house		Experimental house	
	Open space	Water	Room	Water	Room	Water
I	43.00	30.00	35.00	26.20	35.00	24.10
II	41.00	31.00	35.00	28.20	34.00	26.70
III	39.00	30.00	34.80	28.20	34.00	27.50
Average	41.00	30.33	34.93	27.53	34.33	26.10

In Figure 1.1 and Table 1.1 for the first day, the data presents the average air and water temperatures recorded every 10 minutes. Upon comparison, it is observed that the ventilated house maintains an indoor temperature 0.3°C lower than the regular house, given an external environment temperature of 41°C. Additionally, the water, analogous to a human body, in the ventilated house exhibits a temperature 1.43°C lower than that in the regular house, with the water temperature recorded at 30.33°C in the external environment.

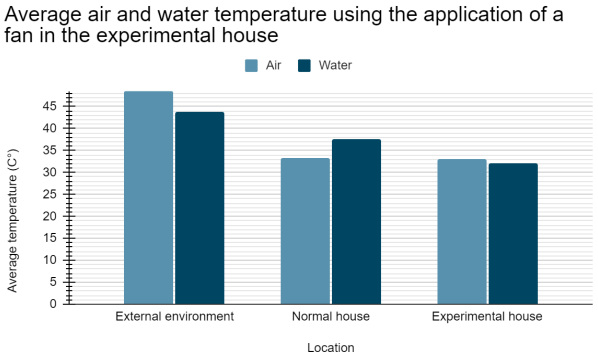


Figure 1.2

Table 1.2 Result of temperatures for experiment 2. Experiment 2

Repetitions	Temperature (C°)					
	External environment		Normal house		Experimental house	
	Open space	Water	Room	Water	Room	Water
I	47.00	43.30	33.00	37.20	33.00	33.40
II	48.00	43.00	33.00	37.50	33.00	29.50
III	50.50	45.00	34.00	38.00	33.50	33.50
Average	48.50	43.77	33.33	37.57	33.17	32.13

Subsequently, Figure 1.2 and Table 1.2 are the results of the 2nd experiment where a fan was applied in the experimental house using the same method. According to the acquired data, the average room temperature in the experiment house is 0.16°C lower compared to that in the regular house, given an external environment temperature of 48.5°C. Finally, the water, representing the human body in the ventilated house, is noted to have an average temperature 5.44°C lower compared to the room temperature in the regular house, with an external environment temperature of 43.77°C.

In this experiment, there are a couple of factors that may affect the results, which includes:

(1) Water beaker: Although the resulting room temperature in both house designs are similar, the air circulation causes a noticeable decrease in the water temperature. This happens because when air passes through water, the air will carry heat from the water and then evaporates, therefore pulling colder air to fall to the bottom whilst discarding the hotter air.

(2) Structure: The independent variable house is designed to be an eight sided (octagon) shape to increase the effectiveness of air circulation. This happens due to the wider angled corners (inside) of an octagon compared to corners of a square/rectangle, which makes the design aerodynamic.

(3) Angle of the walls: Each piece of bamboo which makes up the walls for the experimental design are tilted around 30º upwards to keep out the heat by sunlight while supporting the air flow.

(4) Fan: The independent variable house design has a fan that aids in consistently optimising air circulation inside the house.

(5) Bamboo material: Bamboo is proven to have the richest amount of silica compared to other plants. Due to the richness in silica and its cavity structure, bamboo is also considered a good thermal insulator.

(6) White paint: Both buildings are painted white due to the phenomenon known as the albedo effect. This phenomenon states that lighter color tend to reflect sunlight as opposed to darker colors which absorb sunlight.

4. Conclusion and Future works

Dynamic air circulation doesn't have a substantial change in terms of air temperature but has an recognisable impact on the water temperature. The water bodies within the house represents a human due to the fact that 70% of the human body consists of fluids. The temperature of the water decreases due to the movement of air circulation, consequently affecting the ambient temperature generated by the water and causing it to be exposed to ordinary temperature air, therefore lowering the water temperature.

While conducting this experiment, there are some things that could've been done better to improve the experiment's result for our future works. Regarding materials, attention should be given to their features as well as their material effectiveness. Furthermore, there should be developments in terms of the shape of the experimental house (recommended to make a cylinder house to optimise the aerodynamics of the subsonic flow of air) While monitoring the temperature, it is important that the thermometer is functioning correctly and is appropriate for the respective room and water temperature. When it comes to fans, consideration should be given to the type of fan being used. It is important to verify that the appropriate fan is being used, along with batteries that have a longer lifespan.

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